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- (4)

- $$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3+ax)^6 \quad (3)$$

$$f(x) = 6x^3 + 17x^2 + 4x - 12$$

(a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$.

(2)

(b) Hence, using algebra, write $f(x)$ as a product of three linear factors.

(4)

(c) Solve, for $\frac{\pi}{2} < \theta < \pi$, the equation

$$6 \tan^3 \theta + 17 \tan^2 \theta + 4 \tan \theta - 12 = 0$$

giving your answers to 3 significant figures.

(2)





Solutions relying on calculator technology are not acceptable.

The number of bees in the colony at the start of the study was 30 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

(a) find the value of p , giving your answer to 2 decimal places.

(3)

(b) Find, showing all steps in your working, the smallest integer value of N .

(5)

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- (3)



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- (4)

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$$(b) \sum_{n=1}^{59} u_n \tag{3}$$

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

- (b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$

(5)



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